

Question 3 — Design, Covariance and Gram Matrices:

Cleaning, Principal Component Analysis, and Clustering

SIMC 2026 Endeavour Challenge (Parts a–e)

Abstract

We analyse a design matrix of $N = 1000$ infrared spectra at $M = 3401$ frequencies. Thresholding the row-norm distribution at its spectral gaps removes 40 empty vials and 36 high-norm outliers, leaving $\tilde{N} = 924$ spectra. The covariance matrix $C = \frac{1}{\tilde{N}} \tilde{X}^\top \tilde{X}$ has total variance $\text{tr } C \approx 7.90$, dominance ratio $\lambda_1/\lambda_2 \approx 3.53$, and a sharp elbow at $k^* = 8$ signal eigenvalues; the “95% of variance” rule is shown to be misleading (it needs 667 components). The leading eigenvector is an interpretable difference-spectrum, and projection onto the top four eigenvectors resolves the data into ≈ 25 chemical concoctions.

1 Setup

$X \in \mathbb{R}^{N \times M}$ has rows = *samples* (vials) and columns = *features* (frequencies); X_{ij} is the absorbance of sample i at frequency j . The supplied array has shape 3401×1000 : it is stored *transposed*, and its feature axis is 3401, not the 3041 quoted in the prompt (a digit transposition). We transpose so rows index samples and use $M = 3401$. The *Euclidean norm* $\|x\| = (\sum_j x_j^2)^{1/2}$ summarises a spectrum’s overall magnitude.

2 Part (a): cleaning

Empty vials record only background noise: small, flat, hence small norm. Sorting the 1000 norms reveals a clean gap — 40 vials at $\|x\| \approx 1.6$, then a void until the bulk at $\|x\| \approx 4.3$ — so we threshold at the gap midpoint (a robust, constant-free choice). This flags exactly $T = 40$ empties (rows 960–999). The upper tail holds a *second* error: 36 vials at $\|x\| \approx 8.9$ (rows 924–959), detached by a gap $\approx 5\times$ wider than any internal gap. We retain i iff $\tau_{\text{lo}} \leq \|x_i\| \leq \tau_{\text{hi}}$ via a Boolean keep-mask (non-destructive, composable, index-preserving), leaving $\tilde{N} = 924$ spectra. Validation: removed spectra are flat while retained ones are peaked, and empties are 50.2% exact zeros versus 24.5% (background noise clipped at zero is exactly 0 with probability $\approx \frac{1}{2}$). Section ?? shows the 36 outliers form one isolated cluster whose removal drops the cluster count by exactly one, so the decision is clean (and reversible).

3 Part (b): variance and the covariance matrix

Mean-centering, $\tilde{X}_{ij} = X_{ij} - \bar{X}_j$ with $\bar{X}_j = \frac{1}{N} \sum_i X_{ij}$, removes the shared mean spectrum so the analysis describes how samples *differ*. The covariance matrix is $C = \frac{1}{\tilde{N}} \tilde{X}^\top \tilde{X} \in \mathbb{R}^{M \times M}$.

Proposition 1. C is symmetric and $C_{jj} = \frac{1}{\tilde{N}} \sum_k \tilde{X}_{kj}^2 = \text{Var}_j$, the variance of feature j .

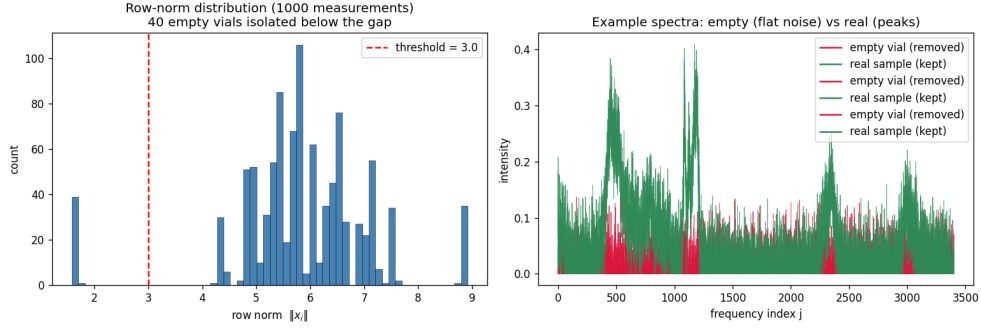


Figure 1: Row-norm distribution: 40 empty vials below the gap; example empty (flat) vs genuine (peaked) spectra.

Proof. $C^T = \frac{1}{N}(\tilde{X}^T \tilde{X})^T = C$; setting $i = j$ and using zero column-means gives the variance. $\square$

Proposition 2. $\text{tr } C = \sum_j \text{Var}_j$ (total variance) is invariant under orthogonal change of basis, and equals $\sum_i \lambda_i$.

Proof. $\text{tr}(Q^T C Q) = \text{tr}(C Q Q^T) = \text{tr } C$ for $Q^T Q = I$; diagonalising $C = Q \Lambda Q^T$ gives $\text{tr } C = \sum_i \lambda_i$. $\square$

Empirically $\text{tr } C \approx 7.90$. Variance concentrates in a few narrow molecular bands (Figure ??); the maximum is at $j^* = 910$ with $C_{j^*j^*} \approx 0.0199$. Most features are flat, so the data is effectively low-dimensional.

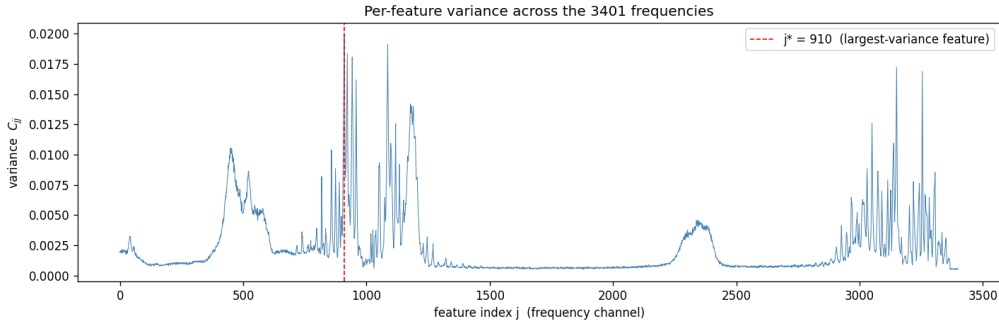


Figure 2: Per-feature variance C_{jj} ; the maximum is at $j^* = 910$ (red).

4 Part (c): the eigenspectrum (PCA)

Proposition 3. For unit w , the variance of the data along w is $w^T C w$; if $C \nu = \lambda \nu$ with $\|\nu\| = 1$ then $\nu^T C \nu = \lambda$. Moreover C is positive semi-definite, so every $\lambda_i \geq 0$.

Proof. The projections $z_i = \tilde{x}_i \cdot w$ have variance $\frac{1}{N} \sum_i z_i^2 = w^T C w$; substituting an eigenvector gives λ . And $w^T C w = \frac{1}{N} \|\tilde{X} w\|^2 \geq 0$, so $\lambda = \nu^T C \nu \geq 0$. $\square$

By the spectral theorem C has real eigenvalues and an orthonormal eigenbasis; we use `numpy.linalg.eigh` (symmetric solver: real eigenvalues, orthonormal eigenvectors, faster and stabler than `eig`), sort decreasingly, and clip rounding-level negatives to 0.

The dominance ratio is $\lambda_1/\lambda_2 \approx 3.53$: one direction dominates, indicating strong low-dimensional structure (a wide spectral gap, the condition for fast Power-Method convergence, Question 2). The

k	1	2	3	4	5	6	7	8	9
λ_k	2.492	0.705	0.590	0.254	0.123	0.105	0.078	0.039	0.0093

Table 1: Leading eigenvalues. The drop $\lambda_8/\lambda_9 \approx 4.1$ is the largest in the spectrum.

elbow is the largest multiplicative drop between consecutive eigenvalues (a ratio, since the signal/noise boundary is multiplicative on the log scale of Figure ??): $\lambda_8/\lambda_9 \approx 4.1$ gives $k^* = 8$. Here λ_8 remains $\approx 4\times$ above the noise floor (signal) while λ_9 lies on it (noise); this matches the ~ 10 dominant molecules. The eight signal components explain only $\approx 55\%$ of the variance, and reaching 95% needs $k = 667$ components (Table ??), because the noise floor is spread over ≈ 900 dimensions; the “95%” rule therefore overstates the dimensionality, and the elbow is the honest measure. Check: $\sum_i \lambda_i \approx \text{tr } C \approx 7.90$ (Prop. ??).

cumulative variance	50%	80%	90%	95%	99%
components k	4	309	516	667	850

Table 2: Components to reach each threshold: contrast the 8 signal components (55%) with 667 for 95%.

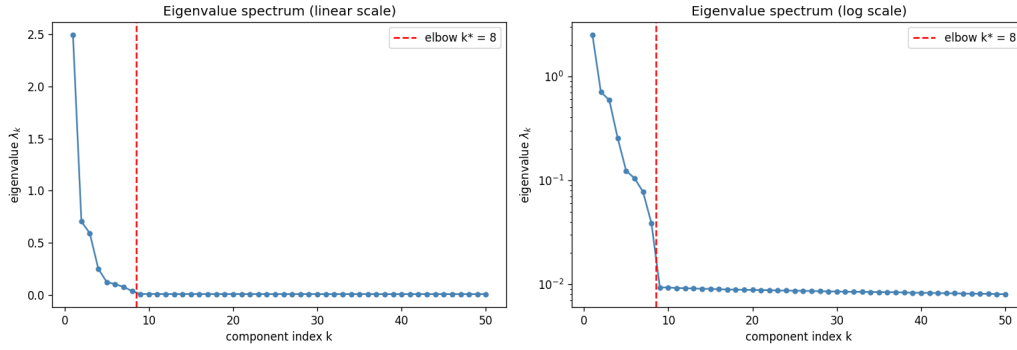


Figure 3: Eigenvalue spectrum (linear and log). The log plot shows a sharp elbow at $k^* = 8$, then a flat noise floor.

5 Part (d): the leading eigenvector

$\nu_1 \in \mathbb{R}^M$ (feature space) gives each frequency a *loading*; the *scores* $z_i = \tilde{x}_i \cdot \nu_1$ are the maximally-spread one-dimensional summaries (Prop. ??). Eigenvectors are sign-ambiguous; we fix the largest loading positive. The loadings are supported on two narrow bands ($j \approx 900\text{--}960, 3150\text{--}3260$) with $\approx 27\%$ negative entries, so ν_1 is a *contrast* (balance of absorption between bands), not over-all intensity. With cosine similarity $\cos(a, b) = \frac{a \cdot b}{\|a\| \|b\|}$ (scale-invariant), ν_1 matches a *centered* spectrum at $\cos \approx 0.92$, but only 0.67 against the raw spectrum and 0.03 against the mean: living in the centered space, ν_1 is $\approx$ orthogonal to the mean and is a *difference-spectrum* (Figure ??), echoing Question 1’s dominant-eigenvector steady state.

6 Part (e): clustering in covariance space

With $V_4 = [\nu_1 \ \nu_2 \ \nu_3 \ \nu_4]$, the compressed coordinates $Z = \tilde{X} V_4 \in \mathbb{R}^{\tilde{N} \times 4}$ give each vial four scores; the $\binom{4}{2} = 6$ pairwise scatters (Figure ??) show many tight, separated groups. Counting by leader

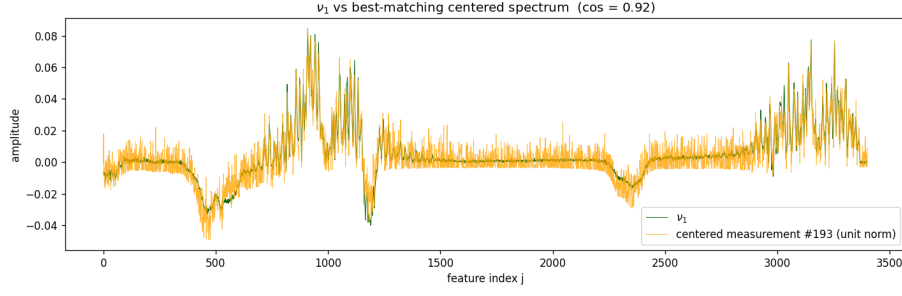


Figure 4: ν_1 (green) overlaid with the best-matching centered spectrum ($\cos \approx 0.92$); a difference-spectrum on two molecular bands.

clustering (a numpy-only threshold method) gives 25 clusters, stable for radii in $[6, 12] \times$ the median nearest-neighbour spacing — a property of the data, not the threshold. Thus the 924 vials form ≈ 25 concoctions, below the “fewer than 100” of Part (h). Although $\nu_1, \dots, \nu_4$ live in feature space, projecting samples onto them exposes sample-space structure, since similar spectra get similar scores. Part (f) repeats this with the Gram matrix $G = \frac{1}{N} \tilde{X} \tilde{X}^\top$ (same nonzero eigenvalues as C , Question 2d), which should recover the same groups.

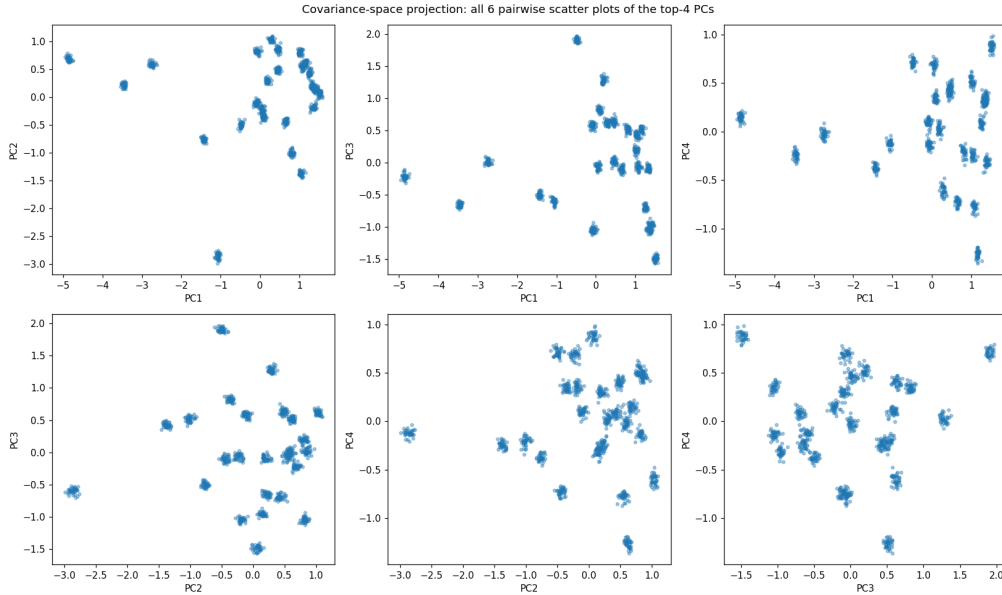


Figure 5: The six pairwise scatters of the top-4 PC scores; each point is one vial. The data resolves into ≈ 25 clusters.